PRST Class (17/09/2020)

09:25

Ex. $\Omega = \{H, T\}$ $X = \begin{cases} 1 & \text{if } H \\ 0 & \text{if } T. \end{cases}$

$P(X=1) = \frac{1}{2}$, $P(X=0) = \frac{1}{2}$.

Discrete rv —
The discrete rv can take only some isolated values.

Ex. $\Omega = \{HHH, HHT, THH, \ldots\ldots, TTT\}$

$$X(\omega) = \begin{cases} 0 & \text{if } \omega = TTT \\ 1 & \omega = TTH,\ HTT,\ THT \\ 2 & \omega = HHT,\ THH\ HTH \\ 3 & \omega = \{HHH\} \end{cases}$$

$P(X=3) = \frac{1}{8}$ $P(X=2) = \frac{3}{8}$

$P(X=0) = \frac{1}{8}$ $P(X=1) = \frac{3}{8}$

X	$P(X=x)$ → probability mass function
0	$\frac{1}{8}$
1	$\frac{3}{8}$
2	$\frac{3}{8}$
3	$\frac{1}{8}$
total	1.

$P(X=x)$: prob mass fn

and $P(X=x) = P(x) \geq 0$ and $\sum_x P(x) = 1$

$S = \{x_1, \ldots\ x_n\}$

~~f(x)~~ $P(x)$ $P(x_i) \geq 0$ $i = 1 \ldots n$

$\sum_{x_i \in S} P(x_i) = 1$

If we have a set $S = \{x_1, \ldots\ x_n\}$ and a function $P(x)$ st

. . n and

If we have " "
and a function $p(x)$ st
$p(x_i) \geq 0 \quad i = 1, \dots n$ and
$\sum_{x_i \in S} p(x_i) = 1$

$p(x)$ is called probability mass function.

Prob dist function:

$F(x) = \sum_{x_i \leq x} p(x_i)$

$F(2.5) = P(X=0) + P(X=1) + P(X=2)$
$= \frac{7}{8}$

$F(0) = P(X=0) = \frac{1}{8}$

$F(-1) = 0$

$F(3) = 1, \quad F(3+0.5) = 1$

Y: # Heads in an outcome + 0.7

Y	$P(Y=y)$
0.7	$P(Y=0.7) = P(X+0.7 = 0.7) = P(X=0) = \frac{1}{8}$
1.7	
2.7	
3.7	

$F_Y(2.5) = P(Y \leq 2.5)$
$= P(Y=0.7) + P(Y=1.7)$
$= P(X=0) + P(X=1).$

Continuous random ~~value~~ variable -
The cont rv can take any value within a certain range.